

Medicine

at Swansea University

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Introduction

This document contains my notes from Swansea University's graduate-entry medicine course. The notes are accessible through GitHub:

<https://github.com/jfbeda/medicine>

You are encouraged to edit and contribute to the notes as you see fit. My contact details are available on my website:

<https://beda.ca/jack>

0.1 Organisation

The main body of the handbook is divided into chapters. Each chapter is stored as a separate file inside the **chapters** folder.

For example:

- `chapters/02-cardiovascular.tex`
- `chapters/03-respiratory.tex`
- `chapters/04-gastrointestinal.tex`

The chapters are imported in the required order from `main.tex`.

CHAPTER 1

Administration

This chapter contains all of the initial administration processes I needed to do, and the order in which I did them.

1.1 Learning@Wales Training

1.1.1 Registering for Learning@Wales

1. Go to learning.nhs.wales.
2. Select **Learning@Wales**.
3. Select the button right below the text that says

‘Please click the button below to register for an account if you are unable to self-register.’

4. Register, and wait to be accepted.

1.1.2 Required training in Learning@Wales

Complete the following training modules:

- [Information Governance, Records Management & Cyber Security](#)
- Fire Safety
- [Infection Prevention & Control Level 1](#)
- Health & Safety
- Treat Me Fairly
- Safeguarding People

- Violence Against Women
- Violence and Aggression
- Manual Handling
- ANTT

1.2 Canvas training

- [NAC - Anti-racism Course](#)
- [NAC - Enhance your EDI Skills](#)

1.3 EOLAS App

- Download the EOLAS app.
- Choose ‘SBUHB Undergraduate Medical Education’ as your organisation.
- Choose ‘SUHB Year 1 Students’ for ‘Is there a team you work most closely with?’

CHAPTER 2

Resources and Useful Links

2.1 Administration links

- [Swansea GEM portal](#)
 - [Student feedback](#)
 - [Leave of absence request](#)
- [Library group study room booking](#)
- [International guidance on paying tuition fees](#)
- [Intranet Student profile](#)

2.2 Learning resources

- [Complete anatomy](#) (activation code: 83548XFMGSQJ)
- [Dissectable Me](#) (5 minute anatomy podcasts)
- [Kenhub](#) (anatomy learning platform)
- [Teach Me Anatomy](#) (anatomy learning platform)
- [Swansea GEM anatomy spotter quiz](#)

CHAPTER 3

Useful Contacts

- **Kenny McKeegan**
- **Carl Rowe**
 - Academic Head of GEM / Programme Director
 - **Office:** SUSIM building?
- **Christian Cobbold**
 - **Office:** Grove building, room 243, near the GEM common room.
- **Brendan Mason**
 - Head of literature review
 - Happy to be emailed again if you haven't received a reply after 48 h
- **Karpagam Krishnamoorthy**
 - **Office:** Grove 154A
- **Lauren Blake**
 - Head of RRHiME
- **Llinos Roberts**

CHAPTER 4

Anatomy

4.1 Anatomical terms

4.1.1 Anatomical planes

- Coronal (frontal) plane: divides the body into anterior and posterior halves.
- Sagittal plane: divides the body into left and right halves. The **median plane** divides the body into equal left and right halves, while a **parasagittal plane** divides the body into unequal left and right halves.
- Transverse (horizontal or axial) plane: divides the body into superior and inferior halves.

4.1.2 Anatomical positional terms

- Superior
- Inferior
- Cephalic (towards the head)¹
- Caudal (towards the tail)¹
- Proximal (closer to the point of attachment)
- Distal (further from the point of attachment)
- Medial
- Lateral
- Posterior/dorsal
- Anterior/ventral

¹The terms **cephalic** and **caudal** are often used in embryology, where the head and tail of the embryo are more distinct than in adults.

4.1.3 Anatomical movement terms

- Supination: rotation of the forearm so that the palm faces anteriorly (or superiorly when the arm is outstretched)
- Pronation: rotation of the forearm so that the palm faces posteriorly (or inferiorly when the arm is outstretched)
- Inversion: rotation of the foot so that the sole faces medially
- Eversion: rotation of the foot so that the sole faces laterally
- Flexion: movement that decreases the angle between two body parts²
- Extension: movement that increases the angle between two body parts²
- Abduction: movement away from the midline of the body
- Adduction: movement towards the midline of the body
- Circumduction: circular movement of a body part, combining flexion, extension, abduction, and adduction
- Rotation: movement around a longitudinal axis, either medially (internal rotation) or laterally (external rotation)
- Elevation: movement of a body part superiorly
- Depression: movement of a body part inferiorly
- Protraction: movement of a body part anteriorly
- Retraction: movement of a body part posteriorly
- Opposition: movement of the thumb to touch the tips of the other fingers
- Reposition: movement of the thumb back to its anatomical position
- Dorsiflexion: movement of the foot so that the toes point superiorly
- Plantar flexion: movement of the foot so that the toes point inferiorly
- Palmar flexion: movement of the hand so that the palm faces inferiorly

² the 'angle' being referred to here is the angle in the 'natural' movement (e.g. at the hip it is the raising of the leg that decreases the angle on the anterior side, which is the side that is usually considered when describing flexion and extension).

4.1.4 Anatomical resources

- [Scrollable axial CT](#)

4.2 Stomach

4.2.1 Cells of the stomach

Cell	Purpose	Location
Gastric parietal cells	Releases gastric acid and intrinsic factor	Fundus and body of the stomach
G-cells	Release gastrin	Primarily in the antrum
D-cells	Produce somatostatin	Stomach/intestine/pancreatic islets
ECL-like cells	Produce histamine	Entire stomach
EC cells	Produce serotonin	Whole digestive tract
P/D1 cells	Produce ghrelin	The fundus
Chief cells	Releases pepsinogen and gastric lipase	The stomach

Table 4.1: Gastrointestinal cell types.

4.2.2 Hormones of the Stomach

Substance	Effect	Released by	Release induced by
Gastrin	↑ HCl release by stimulating parietal cells and stimulating ECL-like cells to release histamine.	G cells	—
Histamine	↑ HCl release by binding to receptors on parietal cells.	ECL-like cells	—
Somatostatin	↓ HCl release and ↓ gastrin release.	D cells	Low pH (high acidity)
Serotonin	Regulates GI motility and fluid secretion.	EC cells	—
Ghrelin	Increases appetite and promotes fat storage.	P/D1 cells	—

Table 4.2: Gastrointestinal signalling substances and their effects.

Note that CCK (cholecystokinin) is released by I cells in the duodenum and jejunum.

CHAPTER 5

Diarrhea

5.1 The Small Intestine

Based on the video *The small intestine: How does the small intestine absorb so much?* by Christian Cobbold.

- Small intestine has many villi and microvilli to increase surface area for absorption.
- Each epithelial cell contains order 3000 microvilli.
- Glucose is absorbed into cells via sodium-glucose transporters (SGLT1) that work by pumping sodium ions out of the cell (towards the interstitial fluid), creating a sodium gradient (with respect to the intestinal fluid), that results in sodium uptake from the intestinal fluid where glucose is co-transported with sodium into the cell.
- Fat globules are emulsified by bile salts which are then (somehow?) turned into micelles (whatever they happen to be?) which are then turned into water-soluble chylomicrons (diameter order 1 μm) that are absorbed into the lymphatic system.

5.2 The Large Intestine

Based on the video *Large intestine: Absorption and defaecation* by Christian Cobbold.

- Age results in decreased gastrointestinal motility, which can result in constipation.
- Age results in decreased large intestine muscle tone and thus straining when defecating, which can result in haemorrhoids.
- Age results in weakening sphincter muscles which results in acid reflux and heartburn.

5.3 Fluid Movement Between Body Compartments

Based on the video *Fluid movement between body compartments: Why doesn't all your fluid stay in the bloodstream?* by Christian Cobbold.

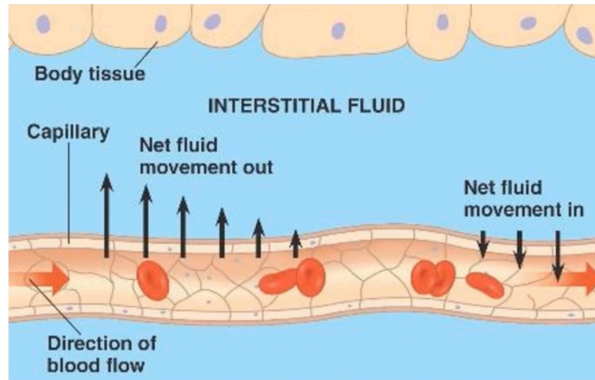


Figure 5.1: Fluid movement between body compartments. [Christian Cobbold]

5.4 Oedema

Based on the video *Oedema: Why fluid accumulates in tissues* by Christian Cobbold.

5.5 2026-09-07 Structure and Function of the GI Tract

By Dr. Christian Cobbold.

CHAPTER 6

Conditions

6.1 Clostridium difficile infection (c. diff)

6.2 Peptic ulcer

Begins with a break in the epithelium which can result from e.g. h. pylori, NSAIDs

30% of ulcers are gastric, and 70% are duodenal [1, 7:32].

6.3 Helicobacter pylori (h. pylori)

Often acquired through oral-oral or oral-fecal in children [1]

h. pylori can activate G cells, which cause the release of more gastrin, and then more gastric acid, which results in mucosal injury [1, t. 5:50].

Gastrin also causes a release of pepsinogen, which turns to pepsin under the influence of acid, which then digests exposed mucosal proteins [1, t. 6:47]. Takeaway message is that pepsin contributes to ulcer deepening, not the cause.

6.4 Gastro-oesophageal reflux disease (GORD/GERD)

GORD is any continuous or regular reflux. Anything that disrupts the lower-oesophageal sphincter can lead to GORD

Long term GORD can cause reflux oesophagitis (mucosal redness and itching). Eventually you can get bleeding and scarring in the oesophagus, and then extremely you can get oesophageal narrowing due to scarring and finally Barrett's oesophagus. [CobboldGastroOesophagealRefluxDisease]

Causes

- Alcohol and caffeine can cause lower-oesophageal relaxation

CHAPTER 7

Drugs

7.1 Antibiotics

7.1.1 Penicillin

7.1.2 Amoxicillin

7.2 Proton Pump Inhibitors (PPIs)

Examples include

- Omeprazole
- Lansoprazole
- Esomeprazole
- Pantoprazole

PPIs travel through the stomach, and are absorbed in the small intestine into the blood. They act on gastric parietal cells to inhibit the release of gastric acid.

Side effects include:

- Increased risk of *c. diff* infection

7.3 Alginates

Examples include

- Gaviscon

- Peptac
- Wind-eze

Forms a 'raft' over the stomach contents.

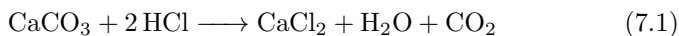
7.4 Antacids

Mechanism of action: neutralizes stomach acid with a chemical reaction.

Examples include:

- Calcium carbonate (CaCO_3)
- Magnesium carbonate (MgCO_3)

For calcium carbonate, the reaction is as follows:



7.5 Prokinetics

Examples include:

- Metoclopramide
- Domperidone
- Erythromycin (off-license)

7.6 H₂-receptor antagonists (H₂RAs)

Mechanism of action: binds to H₂ receptors on gastric parietal cells, preventing histamine from binding and thus reducing gastric acid secretion.

Examples include:

- Famotidine
- Cimetidine

7.7 Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

Examples include:

- Ibuprofen
- Aspirin
- Naproxen
- Diclofenac

Mechanism of action

After injury, arachidonic acid is released from the cells, which is converted to prostaglandins by COX (Cyclooxygenase) which lead to pain. [1, 10:–]

Prostaglandins produce pain/inflammation/fever. If we inhibit COX-2, we can avoid producing prostaglandins, but as a side effect that prevents the protection of the gastric mucosa.

Dangers

NSAID use increases the risk of peptic ulcers as follows [1, 11:–]. In the stomach, prostaglandins are continuously produced from arachidonic acid, and stimulate the production of mucus and bicarbonate, and help with epithelial cell repair. NSAID consumption then inhibits prostaglandin production and has some knock-on effects.

CHAPTER 8

Acronyms

- NGT: Nasogastric Tube (2026-09-11 lecture with Pramodh V)
- ORS: Oral Rehydration Solution (2026-09-11 lecture with Pramodh V)
- LOCS: Learning Opportunities in Clinical Settings
- CC: Cubic Centimeter
- SAM: Severe Acute Malnutrition (2026-09-11 lecture with Pramodh V)
- MUAC: Mid Upper Arm Circumference (2026-09-11 lecture with Pramodh V) (9.8 cm)
- SCM: Sternocleidomastoid
- PC: Presenting Complaint
- HPC: History of presenting complaint
- GORD: Gastro-oesophageal reflux disease
- GERD: Gastro-esophageal reflux disease
- CNS: Central Nervous System
- PNS: Peripheral Nervous System

Bibliography

1. Christian Cobbold. *Peptic ulcer disease: Why stomach ulcers develop* YouTube. Accessed 22 September 2026; cited at 3:37. 2026. <https://www.youtube.com/watch?v=9v8AdFfuVcg&t=217s>.